

Research Statement

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Introduction and Overview

Outline of my research. My primary research focuses on the Representation Theory of Algebras and its rich interactions with Combinatorics and Algebraic Geometry [Mo1, Mo2, Mo3]. Motivated by both foundational results and contemporary developments in these fields (see Section 1), I have developed a long-term research program around a family of open problems that I and others posed over the past decade, nowadays collectively known as the *brick-Brauer–Thrall (bBT) conjectures* (see Section 2). This program is centered on *bricks*, i.e. modules whose endomorphism algebra is a division algebra, and on their role in τ -tilting theory, stability conditions, torsion theory, and the geometry of representation varieties [MP1–MP7]. Over the past year, this multifaceted program has reached a new stage. More specifically, we proved the Second bBT Conjecture for all E -infinite algebras [MP7], showed that every brick-infinite algebra admits infinitely many non- τ -rigid bricks [MP6], and established a lattice-theoretical characterization of brick-directed algebras via left modularity and extremality, for finite as well as infinite lattices of torsion classes [AIMP1, AIMP2, AIMP3].

Beyond my central research agenda, I actively pursue collaborative projects that explore applications of Representation Theory in related disciplines, as in my previous work [B-M+, DMP, KM+]. Furthermore, I have recently developed new interdisciplinary research on emerging connections with Topological Data Analysis (TDA), see Section 3, and I develop computational tools for the research community, see Section 4.

Visibility and impact of my research. According to Google Scholar, as of October 2026, my research manuscripts (11 published articles, 6 preprints under review, and a survey) have received more than 310 citations, 270 of which were received over the past five years, reflecting the interest of the broader mathematical community in my topics of study. Most recent citations pertain to the bBT conjectures, which have proven a fertile ground for original research and have attracted the attention of several research groups. Notably, a recent report on the developments of τ -tilting theory by its founders recognizes this line of research as a notable advance [AIR2, Section 3.2]. Our recent survey [MP5] gives a unified account of bricks across τ -tilting theory, torsion theory, geometric representation theory, and invariant theory. I have also promoted these problems as an invited speaker or organizer at more than 30 international workshops and conferences over the past 6 years.

Content of this document.

- Section 1 recalls the origins of my program: a mutation phenomenon and a modern notion of finiteness that motivated my earlier work and PhD dissertation [Mo1, Mo2, Mo3, MP1].
- Section 2 constitutes the core of this text. More specifically, it describes the bBT conjectures, the recent breakthroughs in [MP6, MP7], the new theory of brick-directed algebras [AIMP1, AIMP2], and my concrete goals for the coming years.
- Section 3 outlines my interdisciplinary research on multiparameter persistence, and Section 4 briefly describes my computational projects and plans for student involvement.

Setting & Notation. Throughout, k is an algebraically closed field and A is a basic, connected, finite dimensional k -algebra. By a theorem of Gabriel, $A = kQ/I$, where Q is a finite connected quiver and I is an admissible ideal of the path algebra kQ . Let $\text{mod} A$ denote the category of finitely generated left A -modules, considered up to isomorphism, and let $|M|$ denote the number of non-isomorphic indecomposable summands of M . We always assume $|A| = n$. For $d \in \mathbb{Z}_{\geq 0}^n$, let $\text{mod}(A, d)$ be the representation variety of modules with dimension vector d , on which $\text{GL}(d)$ acts by conjugation, and let $\text{Irr}(A)$ denote the set of irreducible components of all such varieties. For undefined terminology, see our survey [MP5] and the references therein.

1. Earlier studies & Origins of my long-term project

1.1. Mutation in (τ) -tilting theory. Various mutation phenomena in mathematics and physics, and the conceptual connections between them, have motivated substantial fundamental research. Roughly speaking, one begins with a set of compatible objects and iteratively replaces one object at a time so that compatibility is preserved. This phenomenon underlies tilting theory [AHK], cluster algebras [FZ1], and τ -tilting theory [AIR1]. In classical tilting theory, the mutation graph $\mathcal{H}(\text{tilt}A)$ of basic tilting modules is not necessarily n -regular. This deficiency is elegantly fixed by Adachi–Iyama–Reiten [AIR1]. Recall that $M \in \text{mod}A$ is τ -rigid if $\text{Hom}_A(M, \tau M) = 0$, where τ is the Auslander–Reiten translation, and *support τ -tilting* if it is a τ -tilting module over $A/\langle e \rangle$ for some idempotent e . The mutation graph $\mathcal{H}(s\tau\text{-tilt}A)$ of basic support τ -tilting modules is always n -regular, and τ -tilting theory has since unified deep ideas from a wide range of topics; see [AIR2].

1.2. τ -tilting finite algebras. An algebra is τ -tilting finite if it has only finitely many basic τ -tilting modules, equivalently, if its mutation graph is finite. Representation-finite algebras are τ -tilting finite, but the converse is far from being true. Since τ -tilting finite algebras have finite regular mutation graphs, they are analogues of cluster algebras of finite type [FZ2], and they exhibit rich combinatorial, homological, geometric, and lattice-theoretical structures (see [AIR1, As1, As2, BST, DIJ, DI+, PPP] and the references therein). While a full classification is out of reach, two questions motivated my dissertation [Mo1]:

Q1: For which families of algebras does τ -tilting finiteness imply representation-finiteness?

Q2: How can one systematically decide whether a representation-infinite algebra is τ -tilting (in)finite?

Since every representation-infinite algebra has a minimal representation-infinite (min-rep-inf) quotient, and because τ -tilting infiniteness passes from algebras to their quotients, min-rep-inf algebras are the first family for which these questions are nontrivial. In that regard, the following theorem lists some of my earlier results.

Theorem 1 ([Mo1, Mo2, Mo3]). (1) A min-rep-inf biserial algebra $A = kQ/I$ is τ -tilting infinite if and only if it is gentle; this is the case if and only if (Q, I) is a barbell quiver or A is the path algebra of an acyclic quiver of type $\tilde{A}$.
(2) A min-rep-inf non-distributive algebra $A = kQ/I$ is τ -tilting infinite if and only if Q has a sink or a source.

1.3. Minimal τ -tilting infinite algebras. In my PhD thesis, I introduced the modern analogue of minimality, which provides a systematic framework for modern problems. We say A is *minimal τ -tilting infinite* if it is τ -tilting infinite but each proper quotient is τ -tilting finite. It is known that, unlike τ -tilting finiteness, tilting finiteness is not preserved under quotients. That being the case, the following is highly nontrivial.

Theorem 2 ([Mo3, MP1, MP2]). An algebra is minimal τ -tilting infinite if and only if it is minimal tilting infinite. Every such algebra A of rank n is central, has no projective-injective module and no node, almost all τ -rigid A -modules have projective dimension one, and $\mathcal{H}(\text{tilt}A)$ is infinite and n -regular at almost all vertices. Moreover, a biserial algebra is minimal τ -tilting infinite if and only if it is a generalized barbell algebra or the path algebra of an acyclic quiver of type $\tilde{A}$; in particular, it is gentle.

2. Main objectives: brick-Brauer–Thrall (bBT) Conjectures

Let $\text{brick}(A)$ denote the set of bricks in $\text{mod}A$ (up to isomorphism), also known as Schur representations. An algebra A is said to be *brick-finite* if $\text{brick}(A)$ is a finite set. By the brick– τ -rigid correspondence in [DIJ], A is a τ -tilting finite algebra if and only if it is brick-finite. Thus the minimal τ -tilting infinite algebras of Section 1.3 are precisely the *minimal brick-infinite* algebras, which play a decisive role in what follows.

2.1. My brick-analogue of the Second Brauer–Thrall Conjecture. The celebrated Brauer–Thrall conjectures (1950s) shaped much of modern representation theory. The Second Brauer–Thrall Conjecture, which remained open for over 30 years, asserts that a representation-infinite algebra admits infinitely many dimensions

$d_1 < d_2 < \dots$, each with an infinite family of indecomposables of dimension d_i ; see [Bo] and references therein. A verbatim brick-analogue of this statement is known to be false. However, at the end of my PhD program, I posed the following two conjectures [Mo1, Chapter 6], later published in [Mo2, Mo3], which are still open.

Conjecture (Second bBT, M. 2019). If A is brick-infinite, there exists $d \in \mathbb{Z}_{>0}$ with infinitely many bricks of dimension d .

Conjecture (Rigid-bricks, M. 2019). If almost all bricks over A are rigid, then A is brick-finite.

The Second bBT Conjecture later also appeared in [STV]. In [Mo3, MP1, MP2], we verified it in several cases, including non-distributive min-rep-inf algebras, minimal brick-infinite algebras with infinitely many unfaithful bricks, and all biserial algebras. These results yielded a decisive reduction. More specifically, *it suffices to verify the Second bBT Conjecture for minimal brick-infinite algebras over which almost all bricks are faithful*.

2.2. The family of bBT conjectures. In [MP4] we gave a unified treatment of several fundamental conjectures posed in recent years [CKW, De, En, Mo1, MP2, Pf, STV], all of which, reformulated in terms of bricks, predict the same characterization of brick-infiniteness.

Conjecture (brick-Brauer–Thrall Conjectures). For an algebra A , the following are equivalent:

- (I) A is brick-infinite;
- (II) (*g-vectors*, 2017 [De]) there is a rational ray outside the τ -tilting fan of A ;
- (III) (2nd bBT, 2019 [Mo1, Mo3]) some brick component $\mathcal{L} \in \text{Irr}(A)$ has no dense orbit;
- (IV) (*Semibrick*, 2021 [En]) A admits an infinite semibrick;
- (V) (*Generic brick*, 2022 [MP2]) A admits a generic brick;
- (VI) (*Stability*, 2023 [CKW, Mo3, Pf]) for some weight $\theta \in K_0(\text{proj} A)$, there are infinitely many θ -stable modules of the same dimension.

Our earlier results show that all bBT Conjectures hold for biserial algebras, where in fact a biserial algebra of rank n is brick-infinite if and only if it admits an infinite family of bricks of dimension at most $2n$ [MP2]; that (I), (II), (III), (IV), and (VI) are equivalent for every algebra with a generalized standard Auslander–Reiten component, hence for every algebra with a preprojective or preinjective component [MP4]; and that all bBT Conjectures are *equivalent* for tame algebras, where they reduce to the existence of a homogeneous brick over minimal brick-infinite tame algebras with almost all bricks faithful [MP4, MP5]. We also introduced the framework of Hom-orthogonal modules [MP4], which characterizes τ -tilting finite algebras as conceptual generalizations of local algebras, shows that the Second bBT Conjecture implies the Semibrick Conjecture, and leads to the following elementary version, avoiding the more technical language used in the bBT conjectures.

Conjecture (Hom-orthogonal, 2024 [MP4]). For an algebra A of rank n , the following are equivalent:

- (1) A is brick-infinite;
- (2) $\text{mod} A$ contains an infinite family of pairwise Hom-orthogonal modules of the same dimension;
- (3) $\text{mod} A$ contains $n + 1$ pairwise Hom-orthogonal bricks.

The implications (1) $\Rightarrow$ (2) $\Rightarrow$ (3) hold, and over tame algebras the conjecture is equivalent to all bBT Conjectures.

2.3. Recent breakthroughs. My most recent work with C. Paquette has substantially changed the landscape of these problems. The first result settles a question previously known only for certain families of algebras.

Theorem 3 ([MP6]). *Let A be a finite dimensional algebra. Then A is brick-infinite if and only if A admits infinitely many bricks which are not τ -rigid. Equivalently, A is brick-finite if and only if almost all (meaning all but only finitely many) bricks over A are τ -rigid.*

This proves the τ -analogue of the Rigid-bricks Conjecture [Mo1, Chapter 6] in full generality, and strengthens several of our earlier contributions [MP3, MP4]. For instance, we had earlier shown that if every brick is τ -rigid then A is representation-finite. Since every τ -rigid module is rigid but not conversely, the gap between Theorem 3 and the Rigid-bricks Conjecture is precisely measured by rigid bricks which are not τ -rigid. Motivated by our arguments, we asked whether there exists an algebra admitting an infinite semibrick consisting of pairwise Ext-orthogonal rigid bricks. Our recent studies and communications with other experts in the area led to an affirmative answer to this question, and we aim to study whether every algebra with this property is necessarily strictly wild.

Our second main result is a proof of the Second bBT Conjecture for a large class of algebras, through a careful analysis of τ -regular irreducible components [BS, MP3, Pf] and the stability theory of King [Ki].

Theorem 4 ([MP7]). *If A is an E -infinite algebra, the Second bBT Conjecture holds for A . More precisely,*

- (1) *If there is a rational ray outside the τ -tilting fan of A , then for some $d \in \mathbb{Z}_{>0}$ and some weight θ , A admits an infinite family of θ -stable bricks of dimension d .*
- (2) *From any τ -regular component of A with no dense orbit, one obtains a brick component with no dense orbit whose points in general position are θ -stable for a given weight θ .*

As a consequence of our main result in [MP7], we obtain the following important implications.

Corollary 5. *Let A be a finite dimensional algebra.*

- (1) *A is stably-discrete (also called multiplicity-free) if and only if A is E -finite.*
- (2) *Demonet's (lattice point) Conjecture implies the stable Second bBT Conjecture.*

As discussed in [MP7], these results have several important consequences. More specifically, *the g -vector Conjecture (II) implies the Stability Conjecture (VI)*, and thus the Second bBT Conjecture (III), which itself implies the Semibrick Conjecture (IV). Therefore, for E -infinite algebras, several of the bBT Conjectures are now settled. In particular, my Second bBT Conjecture is reduced to brick-infinite algebras which are E -finite. Combined with our earlier reduction, the open case is thus confined to *minimal brick-infinite, stably-discrete algebras with almost all bricks faithful*, a strikingly restrictive class that I plan to attack in my future work.

2.4. Brick-directed algebras and left modular lattices. In a parallel collaboration with S. Asai, O. Iyama, and C. Paquette, we introduced *brick-splitting torsion pairs* as a modern analogue of the classical splitting torsion pairs. In [AIMP1], we call a torsion pair brick-splitting if every brick is either torsion or torsion-free. Their study leads to *brick-directed algebras*: those admitting no cycle of nonzero non-isomorphisms between bricks; this is a novel generalization of representation-directed algebras, and includes an abundance of examples of tame as well as wild representation type. We gave several characterizations of brick-directed algebras, including via the wall-and-chamber structure [BST, As2] and via a Newton polytope naturally attached to A . In fact, we proved that a brick-finite algebra is brick-directed if and only if its lattice of torsion classes $\text{tors}A$ is left modular, equivalently extremal [AIMP1, AIMP3]. These are fundamental notions for finite lattices [Ma, Th, TW], whose conjunction with semidistributivity characterizes the semidistributive trim lattices. Thus, for lattices of torsion classes, we give a simultaneous characterization of them via representation theory.

As $\text{tors}A$ is infinite for brick-infinite algebras, extending this result required new lattice theory. In [AIMP2], we generalized left modularity and extremality to important families of complete infinite lattices, including the well-separated κ -lattices and weakly atomic completely semidistributive lattices. These families coincide for finite lattices, but differ in general. We introduced new variations of separatedness, and for a large family of complete lattices, including the aforementioned two families, we proved that extremality and left modularity are equivalent, and characterized left modular elements conceptually. Our new work generalizes many known results for finite lattices to a large subfamily of complete lattices, beyond the lattices of torsion classes.

Theorem 6 ([AIMP1, AIMP2]). *For any finite dimensional algebra A , the lattice $\text{tors}A$ is left modular if and only if it is extremal, if and only if A is brick-directed.*

2.5. Infinite-dimensional bricks. Generic bricks, which appear in Conjecture (V), are infinite-dimensional, connecting the bBT program with the theory of generic and endofinite modules [CB, BPS]. With M. Deaconu and C. Paquette, we gave a complete classification of infinite string modules that are bricks over the double-Kronecker gentle algebra, proving that each family is in bijection with a family of Sturmian binary words [Lo], and generalized several of these results to a larger family of gentle algebras [DMP]. This reveals a surprising bridge between bricks and symbolic dynamics and combinatorics on words. This study of infinite-dimensional bricks supplements our earlier work on generic bricks over gentle (and more generally, biserial) algebras, and provides a complementary perspective to recent developments on the existence of generic bricks [BPS, Sc], which are inspired by our open Generic brick Conjecture (V) from [MP2].

2.6. Future goals. Building on the above, my main objectives for the coming years are the following.

- (a) *Completing the Second bBT Conjecture.* Treat the remaining class of minimal brick-infinite, stably-discrete algebras with almost all bricks faithful, combining the geometric techniques of [MP7] with τ -convergence [MP3], and the reduction to homogeneous bricks in the tame case [MP4, MP5].
- (b) *From τ -rigid to rigid.* Regarding the question raised in [MP6], showing that every algebra with an infinite semibrick of Ext-orthogonal rigid bricks is strictly wild would be a major step toward the Rigid-bricks Conjecture.
- (c) *The Hom-orthogonal Conjecture.* Establish (3) $\Rightarrow$ (2) in general, which via [MP4] would yield a new, elementary route to the Semibrick Conjecture.
- (d) *Lattices and brick-directedness.* Classify brick-directed algebras within important families (such as gentle, string, and beyond), and study the distributive sublattice of left modular elements of $\text{tors} A$ and its combinatorial avatars, such as rowmotion [TW] and Newton polytopes [AIMP1].
- (e) *Infinite bricks and generic modules.* Extend the Sturmian classification of [DMP] to all gentle algebras, toward the study of infinite-dimensional bricks. Moreover, as in [CB], relate generic modules and Conjecture (V) to several modern notions of tameness (brick-tame, g -tame, E -tame) [AI, BD, PY].

3. Interdisciplinary research: Representation Theory and TDA

I have recently developed collaborative projects between Representation Theory (RT) and Topological Data Analysis (TDA), with particular focus on the algebraic treatment of Multiparameter Persistence Theory (MPT). Broadly speaking, persistence modules are functors from posets to vector spaces; and many problems in MPT can be formulated in terms of representations of bound quivers [Ou]. It is well known that one-parameter persistence modules are completely classified via representations of type $\mathbb{A}$ quivers. In contrast, multiparameter persistence is significantly more complicated, as it quickly enters the realm of wild representation type [CZ], and many techniques from the easier setting no longer apply. Current research focuses on invariants of persistence modules, and recent interactions between RT and TDA have led to notable advances [BL, BBH, AE+], including combinatorics of gentle algebras [BO+] and geometric representation theory [FJ+]. In ongoing work, we study such modern treatments and in connection with my earlier studies in [B-M+, MP3, MP7].

Spread modules (also known as interval modules) play a pivotal role in the homological treatment of MPT [AE+, BBH], and form a special class of bricks. Inspired by my recent joint work [AIMP1], in an ongoing project, we introduce the family of *spread-directed posets*, namely those which admit no cycle of nonzero non-invertible maps between spread modules. This notion generalizes the brick-directed algebras, and we aim to determine *which (possibly infinite) posets are spread-directed*. Starting from commutative ladders [EH], we have obtained promising results, and we expect that challenging MPT problems become significantly more tractable over spread-directed posets. Beyond my collaborations, to foster this interface between RT and TDA, I led the 7-week thematic program *TDA PARTI* at OIST (2025), and I am the leading organizer of the J-PEAKS workshop *IMPRINT* (Okinawa, February 2027) and the CIRM research school *GRAPE* (Luminy, June 2027).

4. Computational tools and student projects

Through AI-assisted projects, I developed the *FDB Applet*, a self-contained, freely available tool for exploring finite dimensional gentle algebras defined by quivers with relations, which extends the functionality of its predecessor, the FD Applet created by H. Enomoto. Moreover, I have built the *Brick Lodge*, a growing online repository of developments on the bBT conjectures and a space for sharing new results, observations, examples, open problems, and ideas. I plan to extend the applet to compute bricks, τ -tilting fans, brick-directedness, and lattices of torsion classes for broader families of algebras, providing experimental evidence for open conjectures. Many of these problems, notably the combinatorics of string bricks, Sturmian words, and lattices of torsion classes, admit accessible entry points, making them well suited to undergraduate and graduate research projects that combine algebra and combinatorics.

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